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ENGINEERING

A hemispherical latitude-gradient sensor for closed-loop monitoring and management of scoliosis correction

Hehua Zhang^{1†}, Yuyu Gao^{1†}, Xinxin He^{1,2†}, Haoyu Wang¹, Jianpeng Zhang^{1,3}, Weibin Zhu^{1,3}, Yuan Guo¹, Zhenlin Chen^{1,3}, Binbin Zhang^{1,4}, Shengxin Jia¹, Kuanming Yao^{1,5}, Yiming Liu^{1,6}, Xinge Yu^{1,3,7*}

Conservative treatment of scoliosis relies on physiotherapy and bracing, yet both approaches lack tools for quantitative, real-time monitoring of corrective forces. Here, we introduce a hemispherical latitude-gradient (HS-LG) sensor as a previously unknown sensing paradigm for soft, curved-body interfaces, using scoliosis as a clinical exemplar. Leveraging a nonzero Gaussian curvature ($K_G \neq 0$) geometry, the device acts as a spatiotemporal mechanical filter, converting normal pressure into in-plane tensile forces to overcome shear and stress artifacts limiting conventional sensors. Integrated into a wireless platform, the HS-LG sensor enables immediate visualization of spatiotemporal pressure dynamics during therapy. In physiotherapy, bilateral deployment established the first quantitative, closed-loop Schroth regimen. Real-time visual feedback amplified targeted asymmetric breathing by ~40% and decoupled nontargeted regional effort, transforming subjective instruction into data-driven neuromuscular internalization. In bracing, embedded sensors mapped pressures during daily activities, uncovering highly dynamic, posture-specific force redistributions, including transient pressure gradient reversals during ambulation, that challenge conventional static orthotic paradigms. Furthermore, age ($r = 0.738$) and body mass index (BMI; $r = 0.733$) emerged as strong predictors of daily wear compliance in the lumbar cohort, while BMI consistently drove interfacial mechanical loading across both spinal regions (lumbar $r = 0.61$; thoracic $r = 0.77$). Together, these results establish the HS-LG sensor design as a versatile, patient-ready platform that links geometry-driven sensing design to personalized, data-informed scoliosis management, while broadly advancing the development of wearable mechanosensing technologies for soft, curved biological surfaces.

INTRODUCTION

Scoliosis, a three-dimensional (3D) spinal deformity, affects 2 to 3% of adolescents worldwide (1), which can lead to progressive cardiopulmonary dysfunction (2, 3), chronic pain, and reduced quality of life (4). The majority of patients with scoliosis can be treated by non-surgical strategies (5–7), such as physiotherapy based on Schroth method (the most typical adopted physiotherapy) (7, 8) and thoracolumbo-sacral orthoses such as the Boston brace (9, 10). Schroth method uses specialized exercises and breathing techniques to improve the body mechanics for 3D correction, requiring patients to develop precise control over breathing force and rhythm (11, 12). Brace therapy, in contrast, applies targeted pressure to specific regions of the patients' trunk to achieve corrective scoliosis (13, 14).

The therapeutic efficacy of both interventions depends critically on the precise application of corrective forces and adherence (15, 16). Currently, these factors can only be assessed indirectly through therapist expertise, subjective brace fit evaluation, or retrospective

imaging (16, 17). The absence of quantitative, real-time mechanosensing represents a fundamental limitation in nonsurgical scoliosis care, contributing to treatment failure, defined as the inability to prevent curve progression or the eventual need for surgery, that escalate from a baseline of ~25% (13, 18) to 59% in patients with poor adherence (<6 hours/day) (19–21). Notably, evidence-based personalized approaches can reduce this surgical rate to 2% (22), underscoring a notable therapeutic gap that objective monitoring could effectively bridge (22).

Thin, soft pressure sensors have been widely used for wearable electronics for on-body continuous monitoring of health-related mechanical vital signals (23–25) and thus seem the most feasible technology for real-time quantitative assessments of scoliosis therapy (26–28). Therefore, selecting sensitive and long-term stable soft pressure sensors with a sensing range of 0 to 15 N at the skin-soft tissue interface is the key issue for this purpose (14, 20, 21). The current commercially used thin-film strain sensors can provide robust stable properties at the required detection range, but they are not sensitive enough to the pressure of soft interface and also still too stiff to conform to the soft and curved body trunk that would cause artifacts and edge effects during dynamic tests (26, 29). The state-of-art soft sensors have been reported with high sensitivity based on the design of microstructures (30–32), development of new materials (33–36), and implantation of post-signal processing (37–39). Nevertheless, accuracy and stability issues present notable barriers for scoliosis therapy applications, as these sensors struggle to maintain reliable performance at the skin interface encountered in clinical practice (21, 40–42).

¹Department of Biomedical Engineering, City University of Hong Kong, Hong Kong, China. ²Department of Orthopaedics and Traumatology, The Chinese University of Hong Kong, Hong Kong, China. ³Hong Kong Centre for Cerebro-Cardiovascular Health Engineering, Hong Kong, China. ⁴Key Laboratory of Photonic-Electronic Integration and Communication-Sensing Convergence, School of Information Science and Technology, Southwest Jiaotong University, Chengdu, China. ⁵Division of Integrative Systems and Design, The Hong Kong University of Science and Technology, Hong Kong, China. ⁶Macao Institute of Materials Science and Engineering (MIMSE) Faculty of Innovation Engineering, Macau University of Science and Technology, Macao, China. ⁷Institute of Digital Medicine, City University of Hong Kong, Hong Kong, China.

*Corresponding author. Email: xingeyu@cityu.edu.hk

†These authors contributed equally to this work.

Here, we report a hemispherical latitude-gradient (HS-LG) pressure sensor with excellent and ultrarobust sensing performance at soft, curved-body interfaces, with sensitivity of 10 mV/N across a broad practical force range of 0 to 15 N. Leveraging a nonzero Gaussian curvature ($K_G > 0$), this architecture enables three cooperative mechanisms: (i) attenuating high-frequency disturbances via curvature-controlled point impedance, (ii) suppressing shear cross-talk via geometric stiffening against buckling, and (iii) amplifying normal-force signals via membrane strain concentration. By systematically converting normal pressure into in-plane tensile forces, these mechanisms bypass modulus-mismatch penalties to achieve selective, high-fidelity sensing at the dynamic skin interface. Systematic comparisons with commercial thin films, microstructured devices, and a structurally identical planar control validated these advantages, confirming that the HS-LG sensor provides precise and robust pressure detection (43, 44). Integrated into a wireless platform, we demonstrate this sensor's clinical utility by establishing first-in-class quantitative frameworks for both active and passive scoliosis interventions. During Schroth physiotherapy ($n = 22$), bilateral sensor deployment enabled a quantitative, closed-loop training paradigm (8, 11). Real-time visual feedback amplified targeted asymmetric breathing intensity by $\sim 40\%$ and strictly decoupled nontargeted regional effort, highlighting the potential to drive the progressive internalization of complex biomechanics independent of prior training exposure. In Boston brace therapy ($n = 18$), continuous embedded monitoring overturned traditional static orthotic assumptions by mapping highly dynamic, task-dependent pressure heterogeneities (21, 45). We quantified extreme spatial loading during brace donning (peak thoracic pressures of ~ 35 kPa) and uncovered transient pressure gradient reversals during ambulation. Furthermore, we identified patient age ($r = 0.738$, $P < 0.01$) and body mass index (BMI; $r = 0.733$, $P < 0.05$) as robust predictors of daily brace wear compliance in the lumbar region, while BMI consistently drove interfacial mechanical loading across distinct spinal zones (thoracic $r = 0.77$; lumbar $r = 0.61$). Ultimately, this geometry-driven platform bridges the gap between flexible device physics and personalized, data-driven scoliosis management (46), linking device physics to actionable clinical workflows (47, 48).

RESULTS

Design of HS-LG sensors for scoliosis management

Conservative scoliosis treatment encompasses both active physiotherapy, such as the Schroth method, and passive orthotic bracing (Fig. 1A). Despite widespread clinical adoption, both approaches lack real-time force quantification capabilities. Therapists rely on visual assessment to guide asymmetric breathing exercises, while orthosis effectiveness is typically evaluated through static fitting procedures or intermittent imaging studies. The fundamental gap prevents patients from receiving immediate feedback on daily adherence or corrective force distribution, ultimately limiting therapeutic effectiveness and treatment outcomes.

To address this unmet need, we designed an HS-LG sensor for seamless integration into both physiotherapy and bracing workflows (Fig. 1A). In physiotherapy applications, bilateral thoracic deployment enables continuous monitoring of breathing-induced pressures, translating subjective training protocols into closed-loop, feedback-driven learning experiences. For bracing applications, distributed HS-LG sensors embedded at the key thoracic and lumbar contact

sites allow continuous mapping of brace-body interaction during daily activities, capturing spatiotemporal heterogeneity in corrective force transmission. Beyond raw sensing, real-time visualization on mobile device interfaces provides intuitive patient awareness, enhances therapist-patient cooperation, and supports personalized decision-making in orthotic design optimization (Fig. 1A, bottom). Together, these functions position the HS-LG sensor as both a precision measurement tool and an interactive clinical platform that bridges mechanical sensing with evidence-based therapy.

The device consists of a metallic serpentine-patterned network encapsulated within a hemispherical silicone substrate, providing conformal contact with curved skin surfaces. Inspired by the mechanosensory architecture of the human fingertip (fig. S1), this hemispherical elastomer model provides a physical basis for sensing at complex soft-tissue interfaces, which dominate scoliosis management. Figure S2 confirms that the hemispherical soft geometry ensures uniform and reproducible deformation under load at the skin interface. Crucially, rather than relying on complex microstructures, the sensor leverages its nonzero Gaussian curvature to govern mechanical response. This geometric architecture (Fig. 1B) converts vertical pressure into in-plane latitudinal tensile stress, concentrating the sensing strain at midlatitudes. This curvature-mediated stress redistribution enhances sensitivity to low-magnitude forces while inherently suppressing shear interference, a critical advantage over conventional planar architectures. To enable continuous, real-time clinical monitoring, the sensor is integrated with a compact wireless module incorporating signal conditioning, Bluetooth Low Energy (BLE) communication, and inductive charging, providing immediate mobile feedback (Fig. 1B and movies S1 and S2).

Fabricated devices displayed uniform hemispherical morphology with consistent integration of the metallic serpentine network (Fig. 1C), demonstrating excellent fabrication reproducibility. Electrical characterization revealed a monotonic, approximately linear relationship between applied force and relative resistance change ($\Delta R/R_0$). Statistical analyses across 20 independent devices (Fig. 1D) showed narrow performance variation, confirming exceptional device-to-device uniformity. The limit of detection (LOD) was further quantified using a signal-to-noise criterion (fig. S3), defined as $3\sigma_{\text{noise}}/S$, where σ_{noise} is the baseline noise and S is the calibration slope. The HS-LG sensor achieved an LOD of 6.0 mN on rigid interfaces and 16.6 mN on soft, body-mimetic interfaces, while maintaining high linearity ($R^2 \geq 0.998$). These results indicate reliable detection of low-magnitude mechanical stimuli under both ideal and physiologically relevant conditions.

We benchmarked HS-LG sensor against commercial thin film (Tekscan FlexiForce A201) and custom microstructured elastomer sensors (based on previously reported fabrication strategies; see Materials and Methods) during standardized human breathing tasks (four measurement sets per sensor; Fig. 1E). Thin-film devices yielded stable but artifact-prone signals due to edge effects and poor conformity to soft tissue contours. Microstructured sensors produced high sensitivity but unstable, disturbance-prone readouts. In contrast, HS-LG sensor consistently achieved reproducible, low-noise waveforms that accurately captured respiratory dynamics. The successful realization of such dynamic on-body sensing requires overcoming the aforementioned three fundamental challenges through the simultaneous optimization of five quantitative indicators: stress concentration factor for signal amplification; zero detection limit and selectivity for disturbance rejection; and modulus mismatch alongside linearity

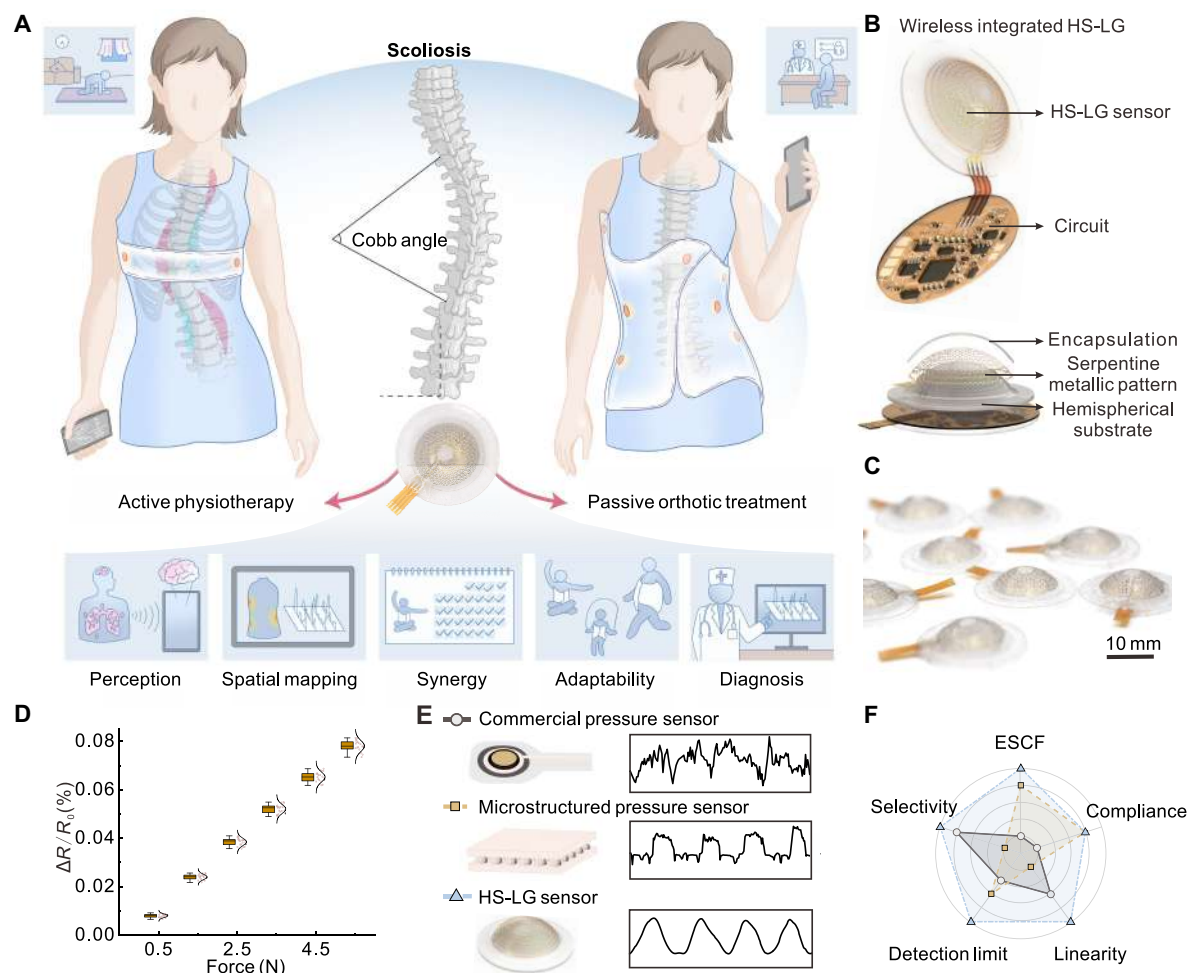


Fig. 1. Application scenario and structural design of the wireless HS-LG pressure sensor for scoliosis management. (A) Conceptual illustration of the sensor enabling real-time feedback during active physiotherapy (mainly Schroth method, for Cobb <20° or ≥20°) and passive orthotic treatment (mainly Boston brace, for Cobb >20°), supporting improved perception, spatiotemporal stress monitoring, synergy, and clinical decision-making. (B) Device structure: wireless integrated HS-LG system with serpentine metallic patterns transferred onto a hemispherical silicone substrate (sensor part) and integrated electronics with receiving coil for wireless energy harvesting (circuit part). (C) Photograph of fabricated HS-LG sensor arrays demonstrating hemispherical geometry and mechanical compliance. (D) Experimental data of the HS-LG sensors, showing the distribution of relative resistance change ($\Delta R/R_0$) versus applied normal force across 20 devices, demonstrating high device-to-device uniformity. (E) Schematic comparison between commercial thin-film sensors, microstructured sensors, and the present HS-LG sensor in terms of output signals, with experimental data of skin-surface pressure variations recorded over four breathing cycles. (F) Radar chart benchmarking key performance metrics: effective stress concentration factor (ESCF), selectivity, resistance to lateral interference, compliance, zero detection limit, and linearity.

for transduction fidelity. These five indicators unfold specifically from the three core aspects to jointly achieve high-fidelity sensing. As summarized in the radar chart (Fig. 1F and note S1), the HS-LG sensor outperforms conventional sensor architectures across all five dimensions, demonstrating the effectiveness of the geometry-driven design strategy for clinical sensing environments. Finite-element modeling (FEM) further demonstrates that HS-LG sensor distributes mechanical stresses more uniformly than planar or microstructured sensors (fig. S4). Scanning electron microscopy (SEM) also confirms robust encapsulation and structural integrity of the sensing network, ensuring reliable long-term sensing performance (fig. S5). Together, the convergence of scalable manufacturing, modulus-matched mechanical compliance, and curvature-driven shear suppression establishes the HS-LG sensor as a robust platform for high-precision biomechanical monitoring in scoliosis management and beyond.

Characterizations of HS-LG sensors

As illustrated in Fig. 2A, the HS-LG sensor comprises stretch-sensitive metallic serpentes encapsulated in a high-modulus styrene-ethylene-butylene-styrene (SEBS) film over a compliant Ecoflex hemispherical substrate. Unlike planar devices where membrane and flexural deformations are mechanically decoupled, the nonzero Gaussian curvature ($K_G \neq 0$) for the hemispherical shell intrinsically couples these modes, enabling it to function as a robust spatiotemporal mechanical filter. Under normal compression, this curvature-mediated coupling converts out-of-plane pressure into distributed in-plane circumferential tensile strain, which is preferentially localized along the latitude-aligned sensing serpentes. This process transforms localized and mechanically unstable indentation into spatially regulated membrane stretching, thereby improving force transduction efficiency and signal reproducibility. The rigorous theoretical derivation and mechanical

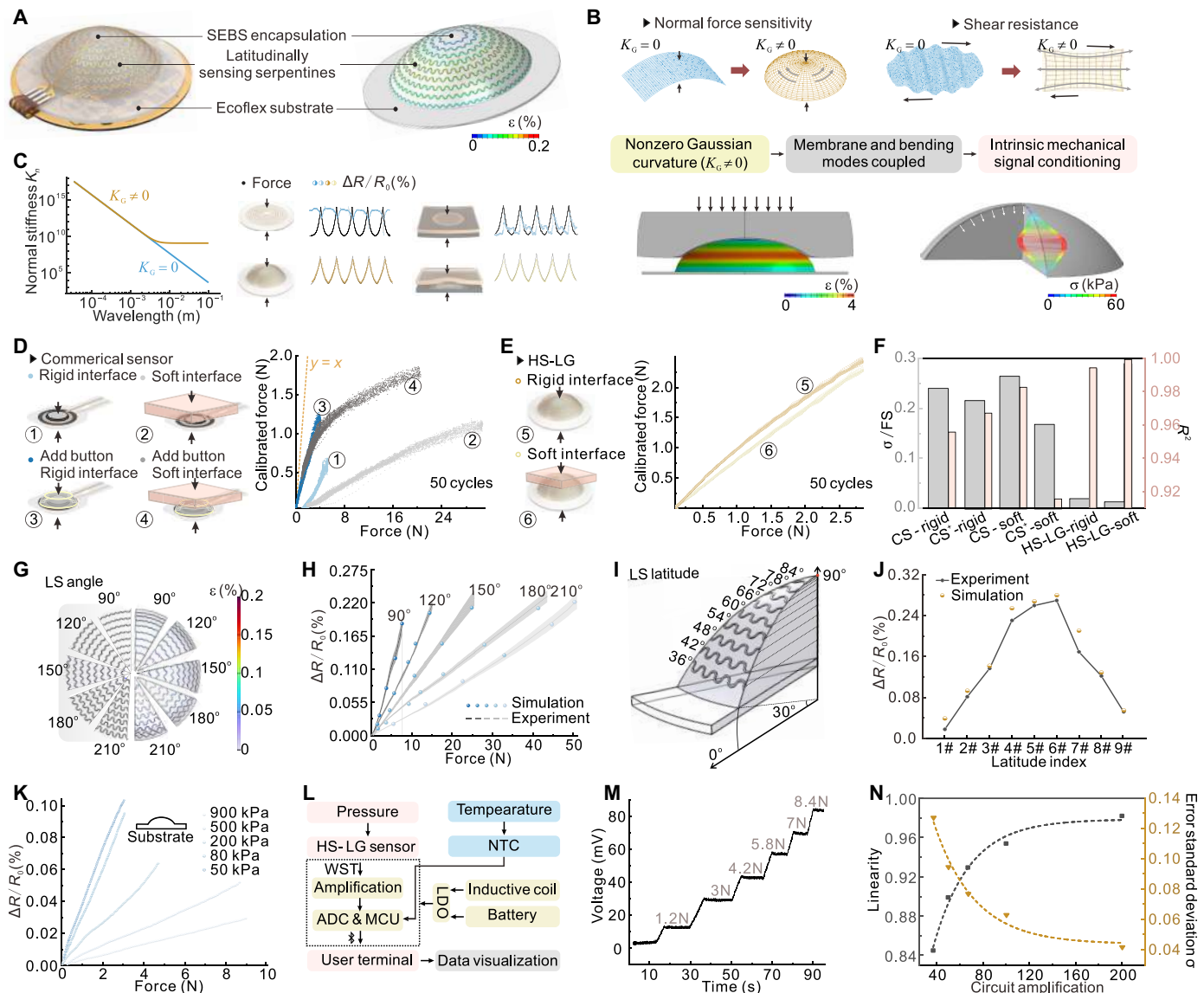


Fig. 2. Sensing mechanism and performance evaluation of the HS-LG sensor against thin-film sensors. (A) Structure and operating principle of the HS-LG sensor. Left: Schematic showing SEBS encapsulation, sensing serpentine, and Ecoflex hemispherical substrate. Right: FEA-simulated strain distribution of latitudinally sensing serpentine under compression. (B) Geometry-enabled force transduction and shear suppression. Top: Comparison of deformation behaviors between planar ($K_G = 0$) and hemispherical ($K_G \neq 0$) structures under normal and shear loading. Bottom: FEA results showing strain distribution and latitudinal tensile force vectors during soft-interface compression. (C) Influence of Gaussian curvature on mechanical cross-talk and interface response. Left: Normal stiffness versus cross-talk wavelength. Right: Responses of identical sensing patterns on planar and hemispherical substrates under rigid and soft interfaces. (D and E) Performance comparison between commercial planar thin-film sensors and the HS-LG sensor under rigid/soft interfaces and silicone-button conditions across 50 cyclic loading tests. Commercial sensors exhibited pronounced variability, nonlinearity, and drift, whereas the HS-LG sensor showed stable and monotonic $\Delta R/R_0$ responses with high repeatability. (F) Quantitative comparison of normalized error (σ/FS) and fitting accuracy (R^2). (G and H) Angular characterization of LS. Polar plots and force-response curves demonstrate that larger angular spans reduce sensitivity while extending dynamic range. (I and J) Latitude-dependent responses of concentric sensing rings distributed from 30° to 90°, showing maximum sensitivity at midlatitudes. Final designs balanced sensitivity and uniform spatial coverage through angle-latitude co-optimization. (K) Dependence of $\Delta R/R_0$ on applied force for substrates with different moduli. (L) Integrated HS-LG sensing system architecture, including sensing bridge, amplification, signal conditioning, wireless communication, and power management. WST, Wheatstone bridge; NTC, negative temperature coefficient thermistor; ADC, analog-to-digital converter; MCU, microcontroller unit; LDO, low-dropout regulator. (M and N) Stepwise loading tests and circuit amplification analysis demonstrate stable, drift-free outputs and improved system-level performance.

analysis underlying this curvature-mediated effect are detailed in note S2. Figure 2B provides an explicit mechanistic comparison between $K_G = 0$ and $K_G \neq 0$ structures using schematic deformation maps. The top panel distinguishes two fundamentally different performance metrics. For normal-force sensitivity, the $K_G \neq 0$ geometry enables efficient conversion of vertical compression into circumferential tensile deformation, producing stronger and more distinguishable electrical responses ($\Delta R/R_0$). In contrast, planar ($K_G = 0$) structures largely dissipate applied loads through local bending, resulting in weaker and less stable signal transduction. For shear resistance, the curved architecture suppresses the propagation of tangential sliding and interfacial friction into the sensing network, thereby minimizing shear-induced deformation artifacts and motion-related noise. The deformation-field comparison highlights that $K_G \neq 0$ simultaneously enhances force transduction efficiency while improving robustness against shear perturbations. Finite element analysis (FEA) further corroborates the geometric phenomenon: Under soft skin-interface compression, the $K_G \neq 0$ geometry and the modulus-mismatched architecture collaboratively convert vertical pressure into in-plane latitudinal tensile forces, localizing maximum expansion at midlatitudes (Fig. 2B).

To explicitly isolate and validate this geometric advantage, we fabricated identical serpentine metallic patterns onto both planar and hemispherical substrates and compared their responses under rigid and soft interfaces (Fig. 2C). The results revealed a notable structural contrast. The left panel reveals the relationship between stiffness and cross-talk wavelength for cases of nonzero and zero Gaussian curvature, respectively, especially among millimeter-to-centimeter scale deformation modes under physiological loading. In this regime, the presence of membrane stresses in the $K_G \neq 0$ structure introduces an additional curvature-dependent stiffness that is absent in planar systems. This geometrically induced stiffness stabilizes the mechanical response and suppresses excessive local deformation. Correspondingly, the right panel shows that the HS-LG sensor maintains stronger, cleaner, and more reproducible pressure signals across both soft and rigid interfaces, whereas planar devices exhibit degraded signal quality under compliant loading conditions. (49, 50)

Together, these results establish that nonzero Gaussian curvature is not merely a geometric feature but a governing design principle that decouples normal-force sensitivity from shear-induced interference, enabling simultaneous enhancement of signal fidelity and mechanical robustness in dynamic skin-interface environments.

Building upon this intrinsic structural superiority, we rigorously benchmarked the HS-LG design against commercial planar thin-film sensors under four distinct conditions: planar structure with rigid interface, planar structure with artificial skin interface, button-modified structure with rigid interface, and button-modified structure with artificial interface. Each configuration was tested using manufacturer-provided calibration and pressure conversion formulas across 50 consecutive loading cycles, with calculated pressures plotted as scatter data (Fig. 2D). The results revealed pronounced nonlinearity, baseline drift, and poor reproducibility across most conditions. Notably, configurations incorporating a silicone button exhibited smaller errors but higher absolute pressure values compared to button-free counterparts, indicating that stepped 3D structures effectively channel stress to designated regions and amplify effective pressure signals. Moreover, soft artificial skin interfaces consistently introduced larger measurement errors than rigid interfaces, confirming the critical role of 3D geometry in stabilizing pressure transmission. The intrinsic response of the HS-LG sensor under rigid and soft interfaces is presented

in Fig. 2E, showing stable and monotonic response with high repeatability across 50 loading cycles. Here, the electrical outputs of both the commercial sensor and the HS-LG device were converted into a unified mechanical quantity (calibrated force, in newton) for direct comparison based on their respective calibration relationships. As shown in Fig. 2F, quantitative comparison results demonstrate that the HS-LG sensor dramatically outperformed commercial sensors, achieving at least an order of magnitude reduction in normalized error/full-scale ratio and exhibiting markedly superior linearity (R^2). This advantage persisted even when the commercial sensors were analyzed exclusively with their quasi-linear regimes. Figure S6 further demonstrates the reproducibility advantage of HS-LG sensor over commercial sensors under curved loading. Long-term stability testing confirmed stable cyclic resistance responses of HS-LG sensor over 700 times loading cycles (fig. S7). To assess viscoelastic creep under sustained loading, extended static compression tests were performed with continuous monitoring of $\Delta R/R_0$ (fig. S8). The response follows a single-exponential relaxation ($R^2 = 0.9905$), with a total baseline drift on the order of 10^{-5} , one to two orders of magnitude smaller than the dynamic signal variations reported here.

Accordingly, for the dynamic activities examined (e.g., breathing and ambulation), drift has negligible impact on signal interpretation. For prolonged static conditions, the fitted relaxation function is implemented in the data-processing pipeline for real-time baseline compensation.

To further harness this geometric signal conditioning, we next analyzed angular dependence arising from the concentric latitudinal serpentine (LS) distribution. Five angular segments (90° , 120° , 150° , 180° , and 210°) were investigated (optical images shown in fig. S9), with corresponding experimental and simulation results shown in Fig. 2 (G and H). Larger angular spans produced lower sensitivity but extended the dynamic force range, reflecting a tunable trade-off between sensitivity and operating window (note S3). Detailed simulation results are provided in fig. S10. In parallel, we introduced the concept of latitude design for the hemispherical geometry, distributing nine LS rings between 30° and 90° (Fig. 2I). Both experimental and simulated data of resistance changes across these rings during pressure application revealed that the middle latitudes exhibited the highest sensitivity, while peripheral regions responded less strongly to applied loads (Fig. 2J).

Integrating angular and latitudinal analyses, we devised a hierarchical optimization strategy pairing higher-sensitivity regions with wider angular spans. This approach ensured uniform sensing coverage across the entire hemispherical surface, effectively eliminating measurement blind zones analogous to the “shortest-stave limitation” in barrel construction, guaranteeing consistent pressure detection throughout the physiologically relevant low-pressure regime. Supplementary FEM data and the final latitude-independent design are presented in fig. S11.

To further optimize the hemispherical design, we systematically analyzed substrate modulus effects on sensor performance (Fig. 2K and fig. S12). Increasing substrate modulus extended the operating range but reduced sensitivity, while softer substrates exhibited higher sensitivity with limited load tolerance. A modulus of ~ 50 kPa was identified as optimal, meeting compliance requirements for skin-mounted applications with sufficient sensitivity for physiological pressure detection. Beyond angular design and substrate modulus, curvature governs both sensitivity and LOD. Systematic simulations were performed by varying hemispherical height while keeping the

diameter constant (fig. S13). Increasing curvature enhances the conversion of normal pressure into in-plane tensile strain, thereby increasing the signal-to-noise ratio at low loads and reducing the LOD. However, higher curvature also limits the sensing range due to material elastic limit, revealing a trade-off between sensitivity, LOD, and dynamic range. Considering these factors, a hemispherical height of 4 mm was selected as the optimal design, providing strong strain amplification while maintaining mechanical stability and reliable fabrication.

Building upon the optimized structural design, the integrated system-level circuit framework is shown in Fig. 2L, including the sensing bridge, amplification, signal conditioning, wireless communication, and power management modules. The circuit implementation used a Wheatstone bridge configuration for resistance-to-voltage conversion and subsequent signal conditioning (fig. S14 and note S4). Data transmission protocols are detailed in figs. S15 and S16. Pressure-voltage calibration of the integrated system confirmed accurate transduction within the physiological range (fig. S17).

Stepwise loading experiments demonstrated clear, drift-free voltage plateaus at each pressure increment (Fig. 2M), while dynamic cyclic measurements across various frequencies validated system stability (fig. S18). These results confirm reliable resolution of both transient and sustained pressures, essential capabilities for capturing respiratory fluctuations and brace-induced static loads on body surfaces.

Regarding thermal effects, HS-LG sensor exhibited strictly linear temperature dependence ($R^2 = 0.998$), with thermal fluctuations nearly an order of magnitude smaller than pressure-induced changes, enabling straightforward computational compensation (fig. S19). The integrated temperature module was validated through 24-hour brace monitoring and calibration against commercial sensors, confirming both measurement accuracy and clinical relevance (fig. S20). Battery performance (fig. S21) showed ~480 hours at 0.1 Hz and ~375 hours at 20 Hz (100 mA-hour), with low current (0.21 to 0.28 mA), enabling multiweek continuous operation under typical brace use. Systematic voltage output comparisons established that amplification factors $\geq 100\times$ provide optimal trade-offs between linearity, error suppression, and energy efficiency for long-term wearable monitoring applications (fig. S22 and Fig. 2N).

Clinical evaluation of HS-LG sensors for Schroth breathing training

We next assessed the translational utility of the HS-LG sensor in scoliosis physiotherapy, focusing on the Schroth method, which redistributes thoracic forces and alleviates spinal asymmetry through asymmetric breathing maneuvers (Fig. 3A) (21). To achieve bilateral monitoring of thoracic wall motion, two wireless HS-LG sensors were temporarily secured to the thoracic wall using an elastic fabric band, enabling real-time acquisition of breathing-induced pressure signals from both sides of the chest (Fig. 3B and movie S3).

A standardized three-phase protocol was applied to all participants, with each phase comprising ~8 to 10 respiratory cycles (Fig. 3C). The first phase involved upright, natural, and regular breathing, and was used as an internal calibration reference, with signals assumed symmetric between the two sensors to minimize potential confounding effects from sensor placement variations or local tissue modulus differences. The second phase introduced therapist-guided asymmetric breathing without real-time feedback, while the third phase used self-monitored training with direct real-time feedback from the

HS-LG sensor system. Additional postures and corresponding patient testing are demonstrated in figs. S23 and S24. A cohort of 22 adolescents with scoliosis (Cobb angle of 10° to 60° , mixed gender, age, and BMI distribution) participated in the study (Fig. 3D and movie S4).

Representative recordings revealed distinct breathing patterns across the three phases (Fig. 3E). During the first phase, participants performed upright natural breathing with real-time bilateral feedback. This phase was designed to help adolescent subjects develop control over deep and prolonged respiration, a fundamental requirement for effective Schroth training. Signals from both sensors overlapped closely, confirming symmetric breathing patterns, while the overall intensity was relatively high due to sustained deep inhalation and exhalation guidance.

During the second phase, therapist-guided asymmetric breathing was introduced without visual feedback. Under these conditions, participants showed a modest increase in breathing effort on Sensor 1, corresponding to the concave side of the scoliotic spine. However, the overall intensity decreased compared to the first phase, and the asymmetry was less consistent across respiratory cycles. This reduction likely reflects the inherent difficulty of maintaining strong and stable asymmetric breathing when relying solely on external manual guidance without continuous visual feedback.

In the third phase, real-time feedback integration allowed participants to directly monitor and adjust their thoracic motion patterns. Consequently, patients progressively enhanced breathing effort on the concave side while achieving more pronounced and stable asymmetry. Notably, repeated respiratory cycles revealed a clear learning effect: Subjects gradually improved both the magnitude and reproducibility of asymmetric breathing patterns, suggesting a potential pathway toward the data-driven neuromuscular internalization of complex biomechanics, which is critical for long-term physiotherapy success.

The conceptual framework underlying this process is summarized in Fig. 3F, where breathing control emerges from closed-loop interactions between the brain, muscles, and respirations, reinforced by sensor-derived feedback. This framework reflects established neurophysiological mechanisms: Real-time sensory input reduces reliance on abstract therapist instructions by providing immediate, quantifiable cues that directly guide motor output. As a result, subjects can adjust respiratory effort with enhanced precision and consistency.

Group-level quantitative analysis corroborated this mechanism (51). From the second to the third phase, real-time visual feedback amplified targeted asymmetric breathing intensity on the concave side (Sensor 1) by ~40%, while strictly decoupling and suppressing nontargeted regional effort on the convex side (Sensor 2) (Fig. 3, G and H). This preferential enhancement confirms that feedback enabled subjects to allocate greater effort to the targeted side, thereby overcoming the natural tendencies toward bilateral breathing symmetry. The outcomes align with motor learning principles, where external feedback accelerates the acquisition of voluntary, task-specific control (52–54).

Intersensor comparisons further validated these effects. As shown in Fig. 3I, nearly all data points were positioned above the unity line, indicating that the bilateral intensity differences—the asymmetry magnitude—were consistently greater during feedback-guided breathing compared to therapist-guided breathing. This demonstrates that feedback not only increased absolute breathing intensity but also sharpened directional control, improving the selectivity of respiratory force application.

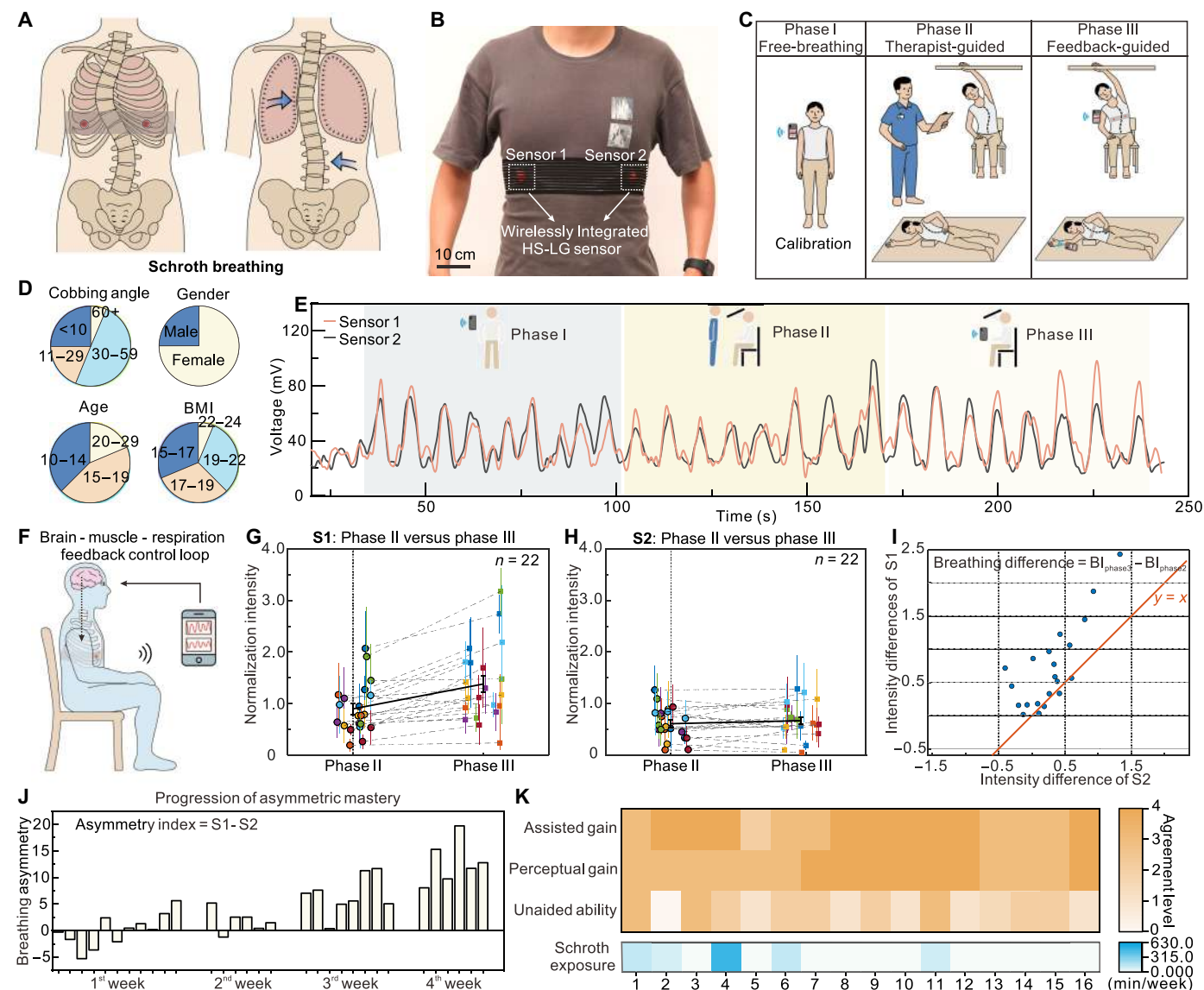


Fig. 3. Application of HS-LG sensor for real-time monitoring and feedback during active physiotherapy—Schroth asymmetric breathing therapy. (A) Principle of Schroth breathing that spinal correction via asymmetrical breathing-induced force redistribution. (B) Dual HS-LG sensors mounted by an elastic fabric band for bilateral breathing monitoring. (C) Standardized protocol consisting of three phases (~8 breaths per phase): (i) baseline symmetric breathing for system calibration; (ii) therapist-guided asymmetric breathing; and (iii) self-monitored training with real-time feedback. (D) Clinical cohort demographics including Cobb angle, age, gender, and BMI. (E) Representative bilateral voltage traces across the three phases. (F) Conceptual framework illustrating the brain-muscle-respiration closed-loop control strengthened by real-time sensor feedback. (G) Normalized intensity comparison for Sensor 1 (S1). (H) Normalized intensity comparison for Sensor 2 (S2). (I) Bilateral intensity differences, with most points above the $y = x$ line, confirming enhanced asymmetry during feedback-guided breathing compared with therapist guidance. (J) Longitudinal tracking of asymmetric breathing mastery in a representative subject over 4 weeks. (K) Heatmap of group-level self-reported agreement, highlighting consistent benefits of real-time visualization and sensor-assisted feedback.

Longitudinal monitoring provided additional validation. In a representative subject, feedback-assisted practice over 4 weeks yielded a progressive increase in both the magnitude and reproducibility of breathing asymmetry, reflecting the motor memory consolidation and improved voluntary exercise mastery (Fig. 3J). Cohort-level subjective evaluations aligned with these findings: Heatmap analysis revealed that participants consistently rated real-time visualization and feedback as essential for perceiving and executing asymmetric breathing, independent of prior training

exposure or exercise experience (Fig. 3K). This coordination between objective measurements and subjective perception highlights the central role of feedback in bridging cognitive awareness and motor execution.

In short, these results establish HS-LG sensor as a clinically practical, wearable, and wireless tool for standardizing and optimizing Schroth-based physiotherapy. By providing bilateral, calibrated, real-time quantification of breathing-induced thoracic forces, the system enhances patient awareness, enables efficient self-supervised

training, and reduces dependence on specialized therapist expertise for effective treatment delivery.

Clinical evaluation of HS-LG sensor for Boston bracing

We used the HS-LG sensor to characterize in-brace pressure redistribution in adolescents with idiopathic scoliosis wearing Boston braces. A schematic of thoracic and lumbar curve locations is provided in Fig. 4A. Pressure sensors were affixed to the inner brace surface using removable adhesive to ensure clinical usability and measurement reproducibility (Fig. 4B). Region-specific calibration

was validated using artificial skin phantoms with layered silicone mimicking thoracic and lumbar tissue mechanics (fig. S25). Eighteen patients were enrolled, representing heterogeneous curve types, Cobb angle severities, BMI categories, and age distributions (Fig. 4C). Notably, the hemispherical architecture forms discrete contact regions with interstitial air gaps, reducing occlusion and enabling passive ventilation. Post-use scores indicated good tolerability (3.47 ± 0.64) with minimal interference (3.27 ± 0.70), and no skin irritation or pressure injury was observed. Furthermore, perspiration in biomedical applications did not induce measurable signal instability, consistent with

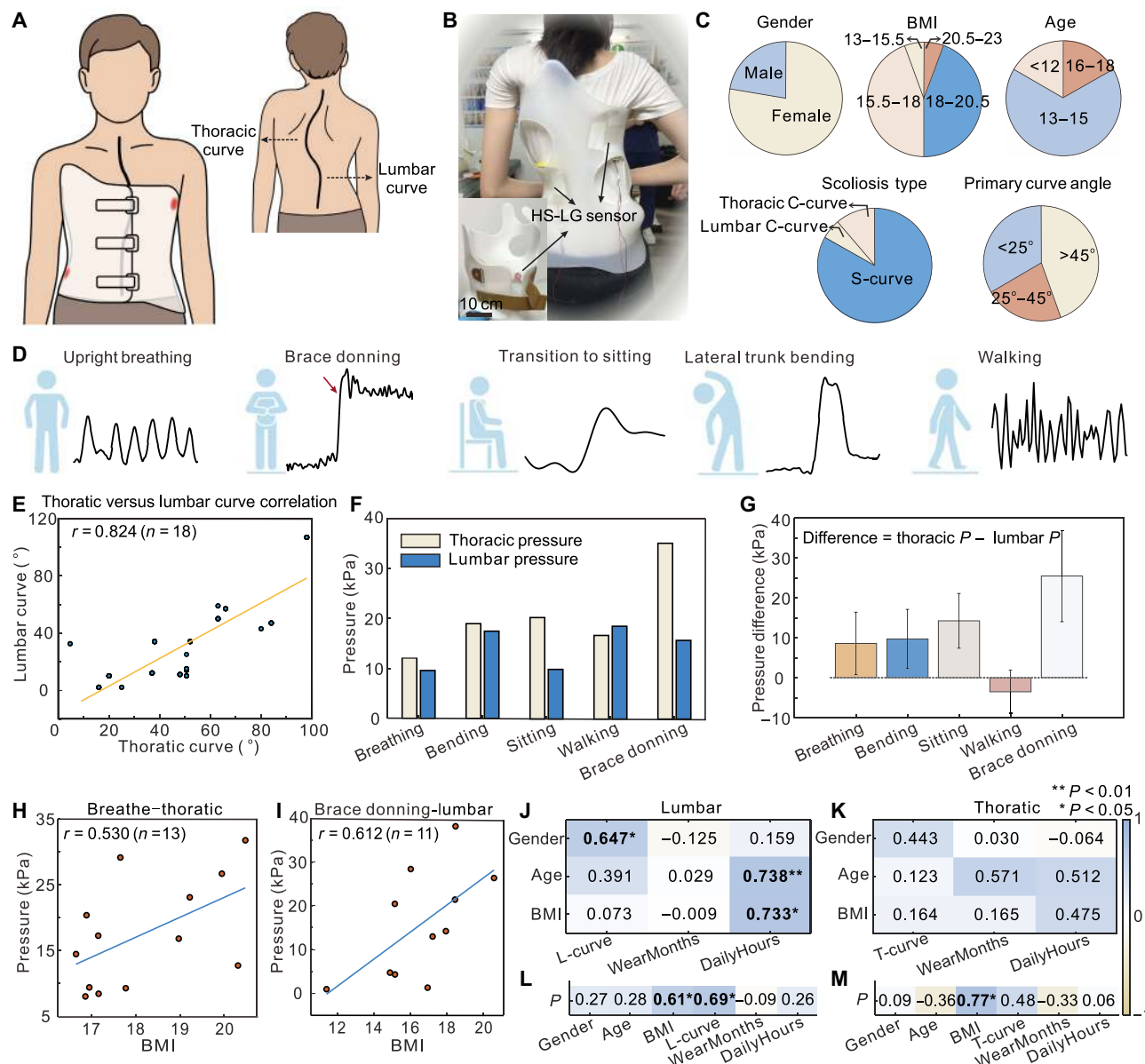


Fig. 4. Quantitative monitoring of pressure distribution during Boston brace treatment using HS-LG sensor. (A) Schematic of thoracic and lumbar curve locations. (B) Patient wearing Boston brace equipped with removable HS-LG sensors (inset: brace with attached sensors). (C) Demographics of the study cohort ($n = 18$) stratified by gender, BMI, age, curve angle, and scoliosis type. (D) Representative pressure waveforms during upright breathing, brace donning, transition to sitting, lateral trunk bending, and walking. (E) Correlation between thoracic and lumbar Cobb angles. (F) Regional pressures across activities. (G) Thoracic-lumbar pressure differences by activity. (H and I) Associations between BMI and pressure in thoracic (breathing) and lumbar (brace donning) regions. (J and K) Correlation heatmaps of clinical factors with brace wear compliance in lumbar and thoracic regions. (L and M) Correlation heatmaps of biomechanical parameters with in-brace pressure differences.

the geometry-mediated transduction based on in-plane tensile deformation rather than interfacial properties.

Dynamic monitoring across five representative daily activities—upright breathing, brace donning, transition to sitting, lateral trunk bending, and walking—demonstrated posture-specific pressure signatures (Fig. 4D). Breathing induced low-amplitude cyclic oscillations, brace donning produced sudden and sustained increases, sitting transition elicited a transient peak, lateral bending generated asymmetric loading patterns, and walking produced rhythmic, high-frequency fluctuations. By mapping these highly dynamic, task-dependent pressure heterogeneities, continuous embedded monitoring fundamentally challenges and overturns traditional static orthotic assumptions. Since the y axis reflects relative variation rather than absolute values, these curves should be interpreted as qualitative trends rather than quantitative comparisons across tasks.

Thoracic and lumbar Cobb angles were strongly correlated ($r = 0.824$, $n = 18$; Fig. 4E), underscoring biomechanical coupling between spinal levels in scoliotic deformity. Activity-specific pressure mapping revealed region-dependent differences, with thoracic pressures generally exceeding lumbar pressures (Fig. 4F). The thoracic-lumbar differential was minimal during breathing and sitting but most pronounced during brace donning, where we quantified extreme spatial loading with peak thoracic pressures reaching ~ 35 kPa. This is consistent with strap tightening concentrating corrective forces at designated pad contact sites (Fig. 4G).

BMI emerged as a significant determinant of pressure distribution patterns. Thoracic pressure during breathing correlated positively with BMI ($r = 0.530$; Fig. 4H), and lumbar pressure during brace donning showed a similar relationship ($r = 0.612$; Fig. 4I). These results support a biomechanical mechanism whereby greater BMI increases soft tissue thickness and compliance, necessitating higher interface forces to effectively transmit corrective loads to underlying skeletal structures. Notably, patients with higher BMI consistently exhibited elevated in-brace pressures across both anatomical regions.

Comprehensive correlational analysis through heatmap visualization provided insights into patient-specific factors and brace-related parameters. In the lumbar region, significant positive associations with daily wear compliance (DailyHours) were observed for both age ($r = 0.738$, $P < 0.01$) and BMI ($r = 0.733$, $P < 0.05$) (Fig. 4J). In addition, gender showed a significant correlation with lumbar curve magnitude ($r = 0.647$, $P < 0.05$). However, neither variable correlated with total wear months, indicating that while daily compliance is influenced by BMI and age factors, brace replacement intervals are governed by independent clinical or behavioral considerations. In the thoracic region, no correlations achieved statistical significance, although moderate correlation coefficients (e.g., age versus wear months, $r = 0.571$; age versus daily wear hours, $r = 0.512$) suggest potential trends that may become evident in larger cohorts (Fig. 4K). These underappreciated findings highlight the variability of thoracic responses.

Biomechanical correlation analysis further indicated that pressure differences during brace donning were significantly associated with BMI in both lumbar ($r = 0.61$, $P < 0.05$) and thoracic ($r = 0.77$, $P < 0.05$) regions, reinforcing BMI as a universal, region-independent modulator of corrective force transmission (Fig. 4, L and M). Furthermore, lumbar curve magnitude correlated significantly with lumbar pressure ($r = 0.69$, $P < 0.05$), whereas thoracic curve magnitude showed only a nonsignificant positive trend ($r = 0.484$), suggesting regional asymmetry in mechanical responsiveness to spinal

deformity severity. Correlations with gender and age were weak and did not achieve statistical significance, although subtle trends (e.g., younger patients exhibiting lower pressure magnitudes) warrant investigation in further studies.

These findings validate the HS-LG sensor as a robust platform for quantifying the complex interplay between anatomical, behavioral, and biomechanical determinants of brace performance. The correlation patterns across Fig. 4 (J to M) and fig. S26 provide actionable clinical guidance: Brace adjustment should prioritize BMI and curve magnitude when calibrating corrective pad forces, whereas compliance-related parameters (e.g., daily wear hours versus brace renewal schedules) should be managed through independent clinical strategies. Recognition of both significant and nonsignificant correlations ensures balanced interpretation and supports refinement of personalized brace prescription strategies in contemporary scoliosis care.

DISCUSSION

This work establishes a geometry-driven sensing paradigm for reliable mechanosensing on soft, curved biological surfaces, with scoliosis management serving as a clinically relevant validation scenario. By introducing a nonzero Gaussian curvature ($K_G \neq 0$) architecture, the HS-LG sensor shifts the design strategy from material-centric compliance matching to geometry-mediated force transduction. Mechanistically, curvature couples bending and membrane modes, enabling conversion of out-of-plane pressure into spatially distributed in-plane tensile strain, which simultaneously (i) amplifies normal-force sensitivity through strain concentration and (ii) suppresses shear-induced artifacts via curvature-dependent geometric stiffening. This decoupling of signal amplification from shear interference constitutes the central advantage over planar ($K_G = 0$) designs.

Systematic benchmarking supports these principles. The HS-LG sensor exhibits high linearity ($R^2 = 0.998$) and an order-of-magnitude reduction in normalized error relative to commercial thin-film devices, while maintaining stable operation across rigid and compliant interfaces. These results indicate that geometry-mediated signal conditioning, rather than microstructural complexity or material modification alone, can govern sensing fidelity during dynamic skin-device interactions. The resulting platform provides robust, low-noise readouts under conditions where conventional planar sensors are limited by modulus mismatch, interfacial shear, and deformation instability.

Clinical translation further demonstrates that these device-level advantages enable actionable physiological measurements. In Schroth physiotherapy, bilateral deployment establishes a quantitative, closed-loop training paradigm. Real-time feedback increases targeted asymmetric breathing by $\sim 40\%$ while suppressing nontargeted effort, indicating improved directional control of respiratory force. Longitudinal observations suggest progressive stabilization of breathing patterns, consistent with feedback-assisted motor learning and neuromuscular internalization. In orthotic management, continuous monitoring reveals highly dynamic, task-dependent pressure heterogeneities, including transient gradient reversals during ambulation and concentrated loading during brace tightening. These findings challenge the conventional assumption of static corrective forces and highlight the need for dynamic, activity-aware brace evaluation.

Correlation analyses further suggest patient-specific determinants of brace performance. Age ($r = 0.738$) and BMI ($r = 0.733$) are associated with daily wear compliance, while BMI correlates

with interfacial mechanical loading across regions (thoracic $r = 0.77$; lumbar $r = 0.61$). These trends are consistent with prior reports linking increased age to improved brace adherence and BMI to variability in orthotic effectiveness and force transmission. However, given the modest cohort size ($n = 18$), these associations should be interpreted as physiologically plausible trends rather than definitive predictors. The observed regional differences—particularly the stronger coupling between lumbar pressure and curve magnitude—further indicate that mechanical responsiveness to correction is not uniform along the spine, supporting the need for region-specific optimization in brace design.

Several limitations remain. First, the sample size in the bracing cohort limits statistical power for correlation analyses. For moderate correlations ($r \approx 0.5$), power analysis indicates that ~30 to 50 subjects would be required for statistically robust clinical validation, exceeding the current cohort size. Accordingly, the observed associations should be interpreted as preliminary but physiologically plausible trends. Second, intersubject variability in soft tissue properties may influence absolute pressure readouts, necessitating improved calibration strategies. Viscoelastic drift under sustained loading is small relative to physiological signals and can be computationally compensated. These considerations define the scope of current conclusions while outlining directions for methodological refinement.

Beyond scoliosis management, the underlying sensing principle is broadly applicable to systems involving soft tissue–device interfaces. The HS-LG platform may be extended to postsurgical monitoring (e.g., spinal fusion recovery) for early detection of abnormal pressure distributions, as well as to rehabilitation scenarios such as orthotic fitting, respiratory training, and motor function retraining. In these contexts, continuous biomechanical quantification combined with real-time feedback may improve adherence and enable data-driven personalization of therapy.

In summary, this work demonstrates that geometry-driven design provides a generalizable route to high-fidelity mechanosensing on soft, curved surfaces. By integrating structural mechanics, wearable electronics, and clinical validation, the HS-LG sensor bridges device physics and actionable health care metrics. This convergence establishes a foundation for feedback-driven, personalized management in scoliosis and related biomechanical applications.

MATERIALS AND METHODS

Fabrication of the encapsulated planar metallic serpentine pattern

Glass substrates were cleaned with acetone, isopropanol, and deionized water and then dried with nitrogen. Polymethyl methacrylate (PMMA) solution (3%, w/v, in anisole) was spin-coated at 3000 rpm for 30 s and baked at 100°C for 15 min to form a release layer. Polyimide (PI) precursor solution was then spin-coated onto the PMMA four times to achieve a total thickness of ~20 μm , with each layer soft-baked at 90°C for 15 min to reach a semidry state before the next deposition. Commercial constantan foil (5- μm thick, Hanzhong Jingce, China) was treated with oxygen plasma for 30 s and mounted on water-soluble tape (WST). The foil was pressed onto the semidry PI layer, and the assembly was immersed in deionized water for 10 min to dissolve the tape. After rinsing and drying, the PI was cured at 250°C for 1 hour in a nitrogen atmosphere. The exposed constantan surface was cleaned with isopropanol. A second PI layer was spin-coated on the opposite side four times and cured at 250°C

for 4 hours in a nitrogen atmosphere. This process obtains a double-sided PI/constantan/PI membrane.

To define the flexible printed circuit (FPC) connection pads, a 100- μm -thick SU8 2050 negative photoresist was spin-coated on the membrane, followed by soft baking at 65°C for 3 min and 95°C for 20 min on a hot plate. The photoresist was then exposed to ultraviolet (UV) light through a mask defining a four-pin layout with 1.0-mm spacing. After postexposure bake and development, the patterned SU8 served as a mask for O_2 reactive ion etching to selectively remove the exposed PI regions, thereby opening the FPC contact ports. The remaining SU8 was removed by overnight acetone soaking.

The encapsulated film was subsequently patterned into open-mesh serpentine architecture with a trace width of 50 μm using a laser cutting system (LPKF ProtoLaser U4). The nonpatterned regions were carefully removed, and the patterned meshes were supported by WST for handling. To form a soft elastomeric encapsulation, SEBS (H1221, 10 wt % in toluene) was spin-coated onto the patterned film. After solvent evaporation in a fume hood for 8 hours, the residual toluene was completely removed. Immersion in deionized water detached the WST and yielded a freestanding SEBS-encapsulated metallic serpentine pattern.

The serpentine mesh layout was designed in a circular geometry but intentionally restricted to a 300° sector rather than a full 360° distribution. This partial configuration was realized by laser cutting to remove the remaining 60° arc of pure SEBS film, which had no patterned traces. The resulting construct constituted the planar metallic serpentine pattern to be subsequently transferred onto the hemispherical elastomeric substrate (55, 56). For comparison studies (Fig. 2), the planar sensors were fabricated with a full 360° circular serpentine distribution, while the hemispherical sensors used the 300° arc design to enable conformal transfer onto the 3D curved surface.

Fabrication of the integrated HS-LG sensor system

The sensor substrate was prepared by casting Ecoflex 0020 (1:1 mixture) into a 3D-printed PLA (polylactic acid) mold, degassing under vacuum for 20 min, and curing at 40°C for 12 hours to obtain a disc-shaped geometry. To reinforce the bottom region serving as the FPC interface, a 100- μm polyethylene terephthalate (PET) ring, laser-cut and bonded with Sil-Poxy, was attached to the circular base. The cured substrate was sequentially cleaned with acetone, isopropanol, and deionized water, dried at 50°C for 2 hours, and plasma-treated. Sil-Poxy was then uniformly applied to the hemispherical surface, onto which SEBS-encapsulated thin films were laminated in 300° fan-shaped segments with aligned radial edges. A custom high-temperature mold was placed over the assembly, followed by thermal pressing at 120°C for 1 min to ensure robust adhesion. Electrical interconnection was achieved by soldering a four-pin flexible FPC cable to the interface, with the joint sealed using silicone. For encapsulation, the device was spin-coated with styrene-butadiene-styrene (D1102, 5 wt % in toluene) containing 0.5 wt % terpene resin and 1 wt % (3-glycidyloxypropyl) trimethoxysilane (Sigma-Aldrich) to enhance coating durability. After solvent evaporation in a fume hood, a second SEBS (5 wt % in toluene) layer was spin-coated and dried under the same conditions. Last, the sensor was connected to the flexible circuit board via the FPC cable, with two power supply options: (i) wireless powering, in which detachable magnetic connectors link the external coil to the circuit, or (ii) battery powering, in which enameled wires were soldered and encapsulated in silicone. For the wireless powering scheme, magnetic induction-based

wireless power transfer technology was used to enable contact-free energy delivery. A pair of copper coils were designed and fabricated as transmitting and receiving coils, respectively. To achieve resonance, a capacitor (47 pF) was connected in parallel with the receiving coil, and the rectification was realized using a Schottky barrier diode (B5819WS) in series, while an additional capacitor was connected in parallel to stabilize the rectified voltage. During operation, the transmitting coil was driven by a signal generator that produced a 3.3-MHz sine wave. When the transmitting and receiving coils were aligned in close proximity, magnetic induction enabled efficient energy transfer, which was then rectified and stabilized to power the entire device.

Fabrication of comparative microstructured elastomer sensor

To benchmark the HS-LG sensor performance, piezoresistive sensors with microstructured architectures were fabricated following established protocols. The device architecture comprised a conductive microfeatured upper layer and a bottom electrode layer. The sensing layer was prepared by uniformly dispersing multiwalled carbon nanotubes (MWCNTs; 5 wt %) into a polydimethylsiloxane (PDMS) precursor (Sylgard 184, 10:1 base-to-curing agent ratio) using ultrasonication to ensure a homogeneous conductive network. The degassed mixture was cast into a custom 3D-printed mold featuring a periodic array of hemispherical microcavities and thermally cured at 80°C for 2 hours. This process yielded a flexible PDMS film with a protruding microsphere array on its surface. For the bottom layer, interdigitated electrodes (IDEs) were fabricated by UV laser patterning of a commercial PI/Cu laminate to form fine-pitch conductive traces. These patterned IDEs were subsequently transferred onto a planar PDMS substrate after surface treatment. The sensor assembly was completed by laminating the MWCNT-PDMS film onto the IDE layer face-to-face. In this configuration, external pressure increases the contact area between the conductive microdomes and the underlying IDEs, resulting in a monotonic decrease in resistance that correlates with the applied load.

Simulation of HS-LG sensor for stress and strain distribution

FEA was performed to simulate the mechanical response of the HS-LG sensor and to optimize its performance. A 3D model was established in Abaqus. The model includes PET substrate, Ecoflex hemisphere module, metal serpentine patterns, SEBS encapsulation, and skin. The Young's moduli of PET, metal, SEBS, and skin are set as 2 GPa, 100 GPa, 1.5 MPa, and 150 kPa, respectively. The Poisson's ratio of PET, metal, SEBS, and skin are set as 0.45, 0.32, 0.45, and 0.45, respectively. Besides, the Ecoflex is set as a superelastic material using Mooney-Rivlin model, where the coefficients C_{10} , C_{01} , and D_1 are set as 1.31 kPa, 0.95 kPa, and -0.93 kPa^{-1} , respectively. The substrate, hemisphere module, encapsulation, and serpentine patterns are tied. The bottom of the substrate is fixed. A C3D8R mesh was applied to the PET substrate and skin, C3D8RH mesh to the Ecoflex, and S4R shell elements to the encapsulation membrane and metal serpentine patterns. The compression process was simulated using the static general module with nonlinear effects enabled.

In addition to mechanical analysis, the resistivity variation of the serpentine patterns under deformation was modeled using COMSOL Multiphysics. The same material parameters were used, combining the solid mechanics and electric currents modules to

couple mechanical strain with electrical resistance change. This allowed quantitative fitting of the piezoresistive response, ensuring consistency between structural deformation and electrical signal output.

Characterization

The current and voltage data were measured by data acquisition and multimeter system (catalog no. DAQ6510, Keithley), with a constant frequency sampling of 1000 Hz. The temperature change was measured by K-type thermocouples (Kaipusen) and infrared camera (catalog no. FLIR A325sc). SEM images were taken using a field-emission SEM (catalog no. LEO-1530). All the pressure tests are performed using the universal testing machine (Instron 5566) with a load cell of 500 N and a testing speed of 15 mm/min. The voltage comparison of the single-arm resistance, the full-bridge differential output, and the amplified bridge output were measured using the PowerLab system.

Integrated HS-LG sensor circuit design

Each pressure sensor has four terminals, incorporating two independent piezoresistive elements arranged side by side. These two elements function as variable resistors in a Wheatstone bridge, where applied pressure induces resistance changes that generate differential voltage signals. The signals are conditioned by a precision instrumentation amplifier (INA333, Texas Instruments) and digitized by the 12-bit analog to digital converter (ADC) of the nRF52810 system-on-chip (Nordic Semiconductor). The same ADC also acquires temperature data from an NTC thermistor.

The nRF52810 serves as both the data acquisition and communication unit, transmitting pressure and temperature signals to a smartphone via BLE. A low-dropout regulator (TPS76933, Texas Instruments) provides stable 3.3-V supply for the system. Dual power modes are supported: wireless charging through a receiver coil and integrated circuit (XKT-R03, XinkeTai Semiconductor) and a thin lithium-ion battery (2 mm). For participant studies, battery operation was primarily used, with silicone encapsulation ensuring flexibility and minimal size increase.

Volunteer and clinical practices

All procedures involving the on-skin attachment of the flexible pressure sensor on volunteers followed ethical guidelines (HUST-A-00001834), which were approved by the Research Ethics Committee of City University of Hong Kong. The clinical study protocol was prospectively registered with the Chinese Clinical Trial Registry (registration number: ChiCTR2600118405). The short-term on-skin attachment tests were carried out by fixing the HS-LG sensor on the brace using medical-grade adhesive tape during daily activities and specified postures. The hemispherical protrusion side of the sensor was oriented toward the skin to ensure direct interface contact, while its soft silicone-based structure provided conformability and minimized the risk of discomfort or skin irritation. Each session lasted ~30 to 60 min under direct supervision to ensure participant safety and comfort. Participants were recruited through open channels in hospitals and the community. Written informed consent was obtained from all participants, and for those under 18 years of age, consent was additionally obtained from their legal guardians. All collected data were anonymized, and safety considerations were strictly followed to minimize risks related to skin irritation or discomfort during sensor attachment.

Clinical expert supervision and evaluation

To ensure clinical relevance and practical validation, all patient experiments in this study were conducted under the direct and continuous supervision of S. W. Law, an orthopedic specialist with over 40 years of experience in scoliosis management and Assistant Dean of the Faculty of Medicine at the Chinese University of Hong Kong, together with orthotist K. W. Ho, a senior orthotic practitioner with more than 40 years of experience in rehabilitation care. K. W. Ho holds professional orthotic certifications from the United Kingdom, the United States, and Canada, and has previously served as an instructor for the International Committee of the Red Cross and as a member of the Educational Committee of the Orthotic and Prosthetic Program in Canada.

Both experts closely supervised all clinical patient experiments and evaluated the resulting performance data. Through structured questionnaires and follow-up interviews, they provided professional assessments and confirmed the effectiveness of the proposed HS-LG sensor system. Specifically, they recognized that the system offered timely data feedback, improved training efficiency, and enhanced patient engagement during Schroth physiotherapy by enabling better recognition and adjustment of breathing patterns. In bracing applications, the sensor was considered valuable for optimizing brace design and improving patient satisfaction by providing quantitative insights into pressure distribution and activity-dependent variations. Overall, both experts acknowledged and endorsed the potential of the HS-LG sensor to advance scoliosis management by transforming subjective training and brace fitting into a data-informed, feedback-driven process.

Supplementary Materials

The PDF file includes:

Supplementary Notes S1 to S4

Figs. S1 to S26

Legends for movies S1 to S4

Other Supplementary Material for this manuscript includes the following:

Movies S1 to S4

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A hemispherical latitude-gradient sensor for closed-loop monitoring and management of scoliosis correction

Hehua Zhang, Yuyu Gao, Xinxin He, Haoyu Wang, Jianpeng Zhang, Weibin Zhu, Yuan Guo, Zhenlin Chen, Binbin Zhang, Shengxin Jia, Kuanming Yao, Yiming Liu, and Xinge Yu

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